Shell Topology Optimization

Shell | Optimization | FEA Engine

INTRODUCTION

Objective

This app helps users generate an efficient shell-type structural layout with variable thickness under uniform distributed loading and self-weight. The goal is to distribute a limited amount of material across the shell so that the final design becomes as stiff as possible. In optimization terms, this is formulated as a compliance minimization problem, where minimizing compliance is equivalent to maximizing global stiffness for a linear elastic structure [1], [2].

The method is based on a variable-thickness topology optimization approach, in which the thickness of each finite element is treated as the design variable. This makes the solution physically interpretable and especially suitable for slab- and shell-like systems, where structural efficiency depends strongly on where thickness is placed [5].

Limitations

This tool is intended for conceptual structural design of shell-like continua. The formulation is based on a linear elastic finite element model, so it is best suited for problems where the response can be reasonably approximated within the small-displacement range [1], [2].

The app idealizes the structure as a continuum shell with equivalent material properties. It does not explicitly model local connections, layered fabrication details, nonlinear material behavior, or failure mechanisms. The results should therefore be interpreted as a design guide for efficient material distribution, not as final construction detailing. The method is also developed for rectangular domains discretized into shell elements, with thickness varying over the surface. While the 3D visualization helps communicate the final geometry, the underlying analysis remains a shell-based optimization model rather than a full solid 3D simulation [6]-[8].

Optimization workflow in the app

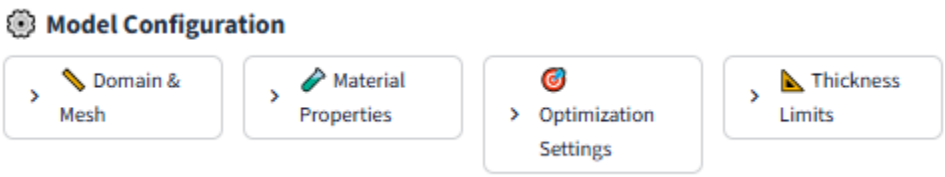
From the user's perspective, the process is straightforward:

1. Define the shell dimensions and mesh.
2. Assign material properties and external loading.
3. Place supports and chooses support types.
4. Set the thickness bounds, material fraction, and filter radius.
5. Run the optimization.
6. Review the evolving thickness map and the final 3D shell geometry.
7. Export the optimized geometry as an STL file if needed.

This workflow turns a classical shell thickness optimization problem into an interactive tool for fast structural exploration.

INPUTS

The app is organized so the user defines the shell geometry, structural properties, support conditions, and optimization settings before running the solver.



Domain and mesh

These inputs determine the geometric resolution of the optimization.

- Domain X (in) and Domain Y (in) define the size of the rectangular shell domain.
- Mesh Size (in) controls the finite element discretization.

Material properties

- Elastic Modulus, E (Pa) controls structural stiffness.
- Poisson's Ratio, v defines the constitutive behavior of the membrane in plane stress.
- Material Density, ρ is used when self-weight is included.
- Distributed Load, w_u represents the uniformly distributed external loading applied to the shell.
- Include Self-Weight adds gravity loading to the model.

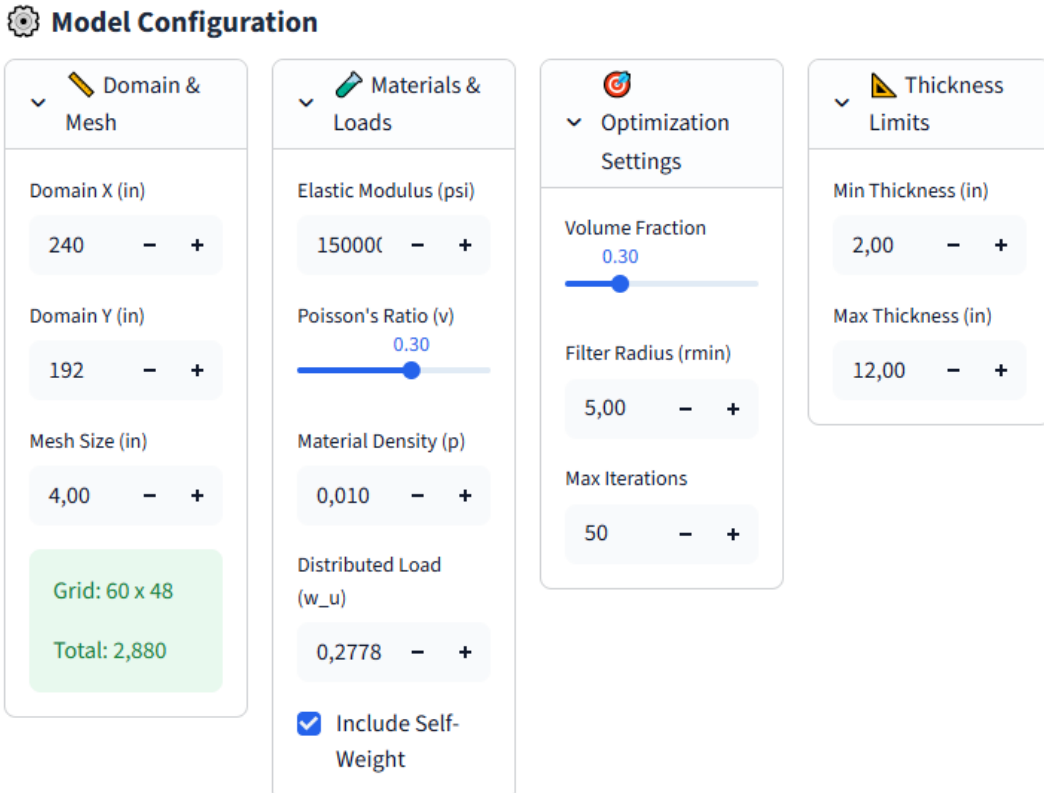
Optimization settings

- Volume Fraction sets the amount of material allowed in the final design. The total volume is given by: $V_0 = V_{frac}(\mathrm{X} \times \mathrm{Y} \times \mathrm{t}_{\min})$
- Filter Radius, rmin regularizes the optimization and helps control numerical artifacts such as checkerboard patterns and mesh dependency, which are common concerns in topology optimization.
- Max Iterations sets the maximum number of optimization steps.

Thickness limits

These bounds define the available design space. In practice, the lower bound can represent a minimum constructive layer, while the upper bound acts as a form-finding limit for the thickest regions of the shell. For this problem no void areas are presented.

- Min Thickness, tmin is the lower thickness bound.
- Max Thickness, tmax is the upper thickness bound.



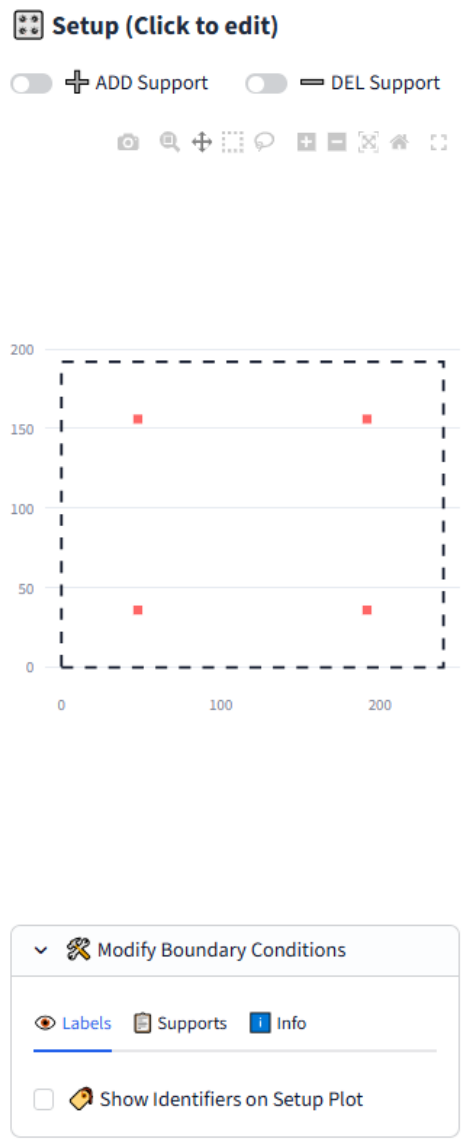
SETUP

Supports

Supports are assigned as rectangular support areas and can be edited graphically or through the support table:

- Pinned support restrains translational movement while allowing rotation.
- Fixed support restrains both translation and rotation.

These support zones define how the shell transfers load to the ground or to adjacent structural components.



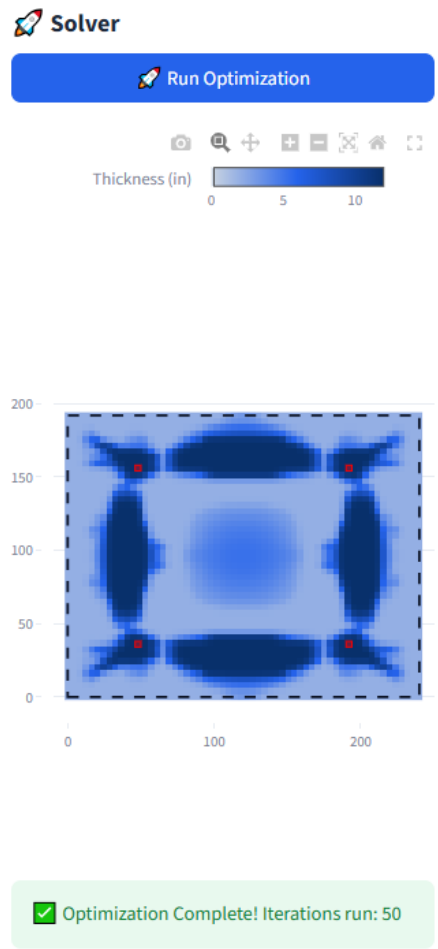
OUTPUTS

Optimized 2D thickness map

The main output is a 2D thickness field over the shell surface. This map shows how material is distributed to improve stiffness:

- thicker areas indicate zones of high structural demand,
- thinner areas indicate regions where less material is needed.

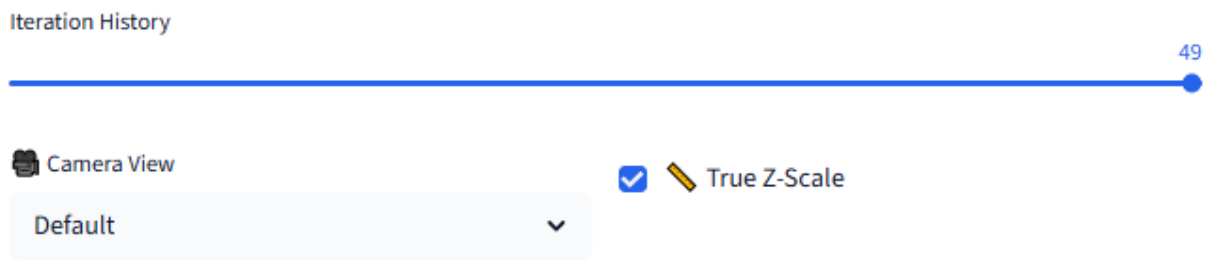
This is the primary result for interpreting the optimized structural behavior.



Iteration history and View options

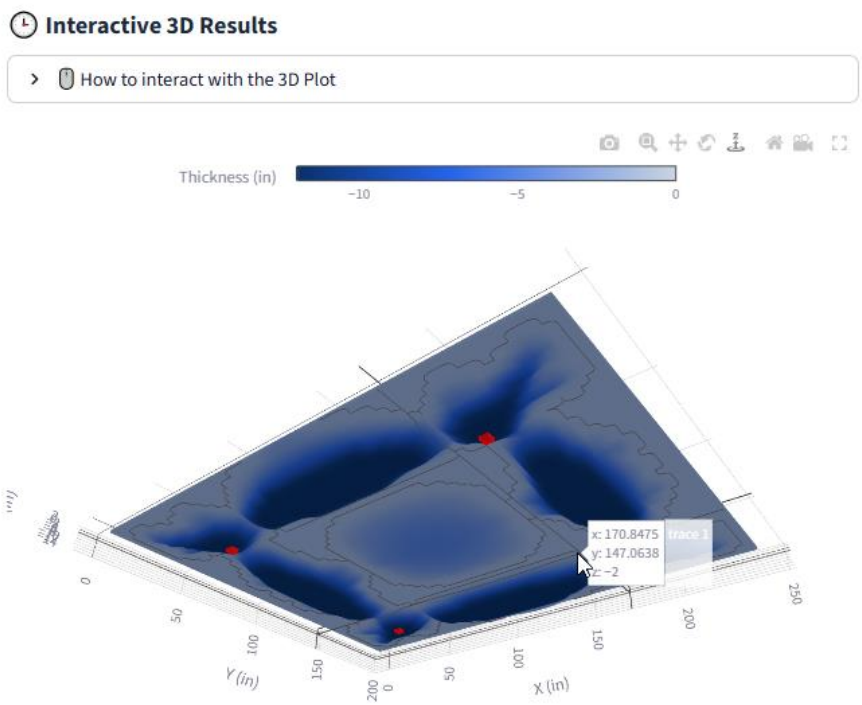
The app stores the optimization history so the user can inspect the progression of the design from its initial state to the final result. This helps reveal:

- how the load path develops,
- whether the design is stabilizing,
- and how material migrates toward the most efficient regions.



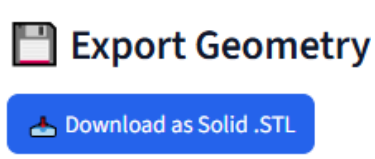
Interactive 3D visualization

The final thickness field is visualized as a 3D solid-like shell surface, making it easier to interpret the design spatially. This representation is especially useful for pavilion, slab, or roof-type structures where thickness variation is part of the architectural expression.



STL export

The final solution can be exported as an STL file, which is useful for presentation models, geometric post-processing, or fabrication-oriented workflows.

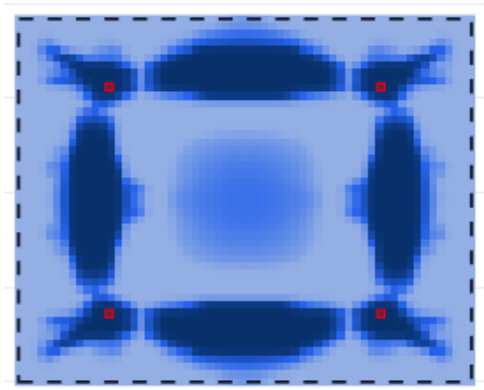


HOW TO USE THE RESULTS

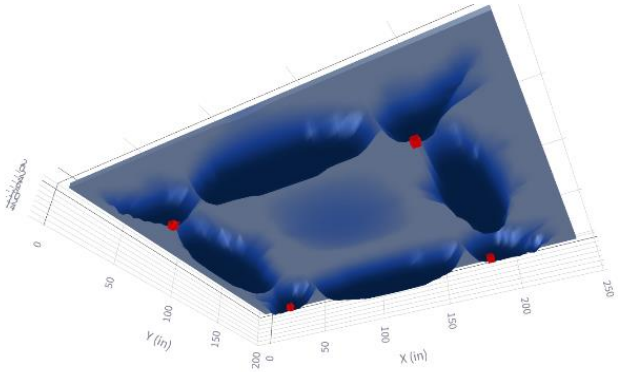
The outputs are best interpreted as a conceptual thickness layout. They help identify:

- the principal load-carrying regions,
- where depth is most valuable,
- where the shell can remain thin,
- and how the structure organizes itself under the applied loads.

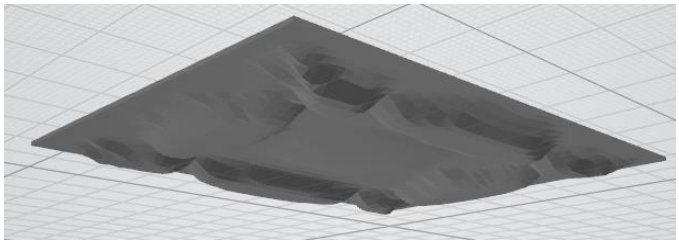
Thickness map



3D Preview



STL



METHODOLOGY

The app follows a variable-thickness shell topology optimization approach inspired by classical compliance-based structural optimization and by thickness-sizing methods for plates and shells [A1], [A5]. The key idea is to treat the thickness of each shell element as the design variable and update that thickness iteratively to achieve the stiffest possible shell for a prescribed amount of material.

Structural idealization

The design domain is idealized as a continuum shell surface and discretized into a mesh of rectangular finite elements. Each element represents a local region of the shell with:

- in-plane dimensions $dx$and dy,
- an assigned thickness t_e,
- and a structural response that includes membrane, shear, and bending effects.

This makes the method suitable for slab- and shell-like systems where thickness variation significantly influences stiffness and load transfer.

Design variable

The optimization variable is the element thickness field

$$\mathrm{t} = [t_1, t_2, \dots, t_{N_e}]$$

where $t_e$is the thickness assigned to element e, bounded by

$$t_{\min} \leq t_e \leq t_{\max}$$

This choice gives the results a direct physical interpretation: instead of deciding whether material exists or not in a purely abstract sense, the method decides how much thickness each part of the shell should receive [A5].

Finite element formulation

Each element is modeled as a Reissner–Mindlin shell element, which captures:

- membrane action,
- transverse shear,
- and bending behavior [A6], [A7].

In the adopted formulation, each node has five degrees of freedom:

- three translations (x, y, z),
- and two rotations $(\theta_{yz}, \theta_{xz})$.

This shell theory is particularly useful because it can represent both thin and moderately thick shells, while avoiding numerical issues such as shear locking when properly implemented [A8].

Objective function: compliance minimization

The objective of the optimization is to minimize the shell compliance:

$$J(\mathrm{t}) = \mathrm{p}^T \mathrm{u}$$

where:

- $\mathrm{p}$is the global load vector,
- $\mathrm{u}$is the global displacement vector.

For a linear elastic structure, compliance is equal to the total internal strain energy and provides a global measure of structural flexibility [A2]. Minimizing compliance therefore leads to the stiffest possible structure for the available amount of material [A1], [A2].

Optimization problem

The shell optimization problem can be stated as:

$$\min_{\mathrm{t}} J(\mathrm{t}) = \mathrm{p}(\mathrm{t})^T \mathrm{u}(\mathrm{t})$$

subject to

$$\begin{aligned} &\mathrm{p}(\mathrm{t}) - \mathrm{K}(\mathrm{t})\mathrm{u}(\mathrm{t}) = 0 \\ &g(\mathrm{t}): V(\mathrm{t}) - V_0 \leq 0 \\ &t_{\min} \leq t_e \leq t_{\max} \end{aligned}$$

where:

- $\mathrm{K}(\mathrm{t})$is the global stiffness matrix,
- $V(\mathrm{t})$is the total material volume,
- $V_0$is the prescribed maximum volume.

This formulation ensures that the final shell remains in equilibrium under the applied loads while using only the allowed amount of material.

Shell stiffness contribution

The stiffness of each shell element depends on its thickness. In the background formulation, the global stiffness is assembled from the element contributions as

$$\mathrm{K}(\mathrm{t}) = \sum_{e=1}^{N_e} \left[t_e \mathrm{K}_{\text{membrane}}^{(e)} + t_e \mathrm{K}_{\text{shear}}^{(e)} + t_e^3 \mathrm{K}_{\text{flexural}}^{(e)} \right]$$

This expression reflects a key physical feature of shell behavior:

- membrane and shear stiffness scale linearly with thickness,
- while bending stiffness scales cubically with thickness.

As a result, even modest thickness increases can produce a strong stiffening effect in bending-dominated regions.

Volume constraint

The total material volume is computed directly from the element areas and thicknesses:

$$V(\mathrm{t}) = \sum_{e=1}^{N_e} dx\, dy\ t_e$$

This means the optimizer does not simply seek the stiffest shell possible; instead, it seeks the stiffest shell for a fixed material budget.

4.8 Load modeling

The global load vector includes two possible contributions:

1. the uniform distributed load applied over the shell surface, and
2. the self-weight, which depends on the material density and the local element thickness.

This is important because the self-weight contribution is not constant: it changes as the thickness field evolves, meaning the loading and the structural stiffness are both linked to the design variable.

Sensitivity-based optimization

To make the optimization computationally efficient, the method uses a gradient-based update strategy [A1]. At each iteration:

1. the shell response is solved using finite element analysis,
2. the compliance is evaluated,
3. the sensitivities of the objective and constraints are computed,
4. the thickness field is updated while enforcing the thickness bounds and volume constraint.

This process is repeated until the iteration limit is reached or the design stabilizes.

Filtering and numerical regularization

Topology optimization problems are prone to numerical artifacts such as checkerboard patterns, mesh dependence, and unrealistically small local features. To mitigate these issues, the method

includes a filter radius $r_{\min}$, which smooths the design field and introduces a practical minimum length scale in the solution [A9].

This filtering step is essential because it improves both the numerical stability of the optimization, and the physical interpretability of the final structural layout.

Why variable thickness is useful for shells

For shell-type structures, thickness is one of the most meaningful design variables because it directly affects:

- in-plane stiffness,
- bending resistance,
- self-weight,
- and the global load path.

A variable-thickness formulation makes it possible to concentrate material in the most structurally effective regions while keeping other areas light. This is particularly attractive in timber shells, slabs, and pavilion-type systems, where depth can be strategically increased to improve stiffness-to-weight performance.

References

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